

GE261004 DESIGN THINKING

Unit I – Introduction to Design Thinking · Presentation 5 of 6

Human-Centred Design and the Engineer's Role

What human-centred means · desirability, feasibility and viability · users, customers and stakeholders · inclusive design · engineering and society

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In this presentation

- What human-centred means
- The three lenses of innovation
- When one lens is missing
- Users, customers and stakeholders
- Inclusive design and extreme users
- Engineering as a social activity
- Where engineers meet society
- Professional and ethical responsibility

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Human-centred design

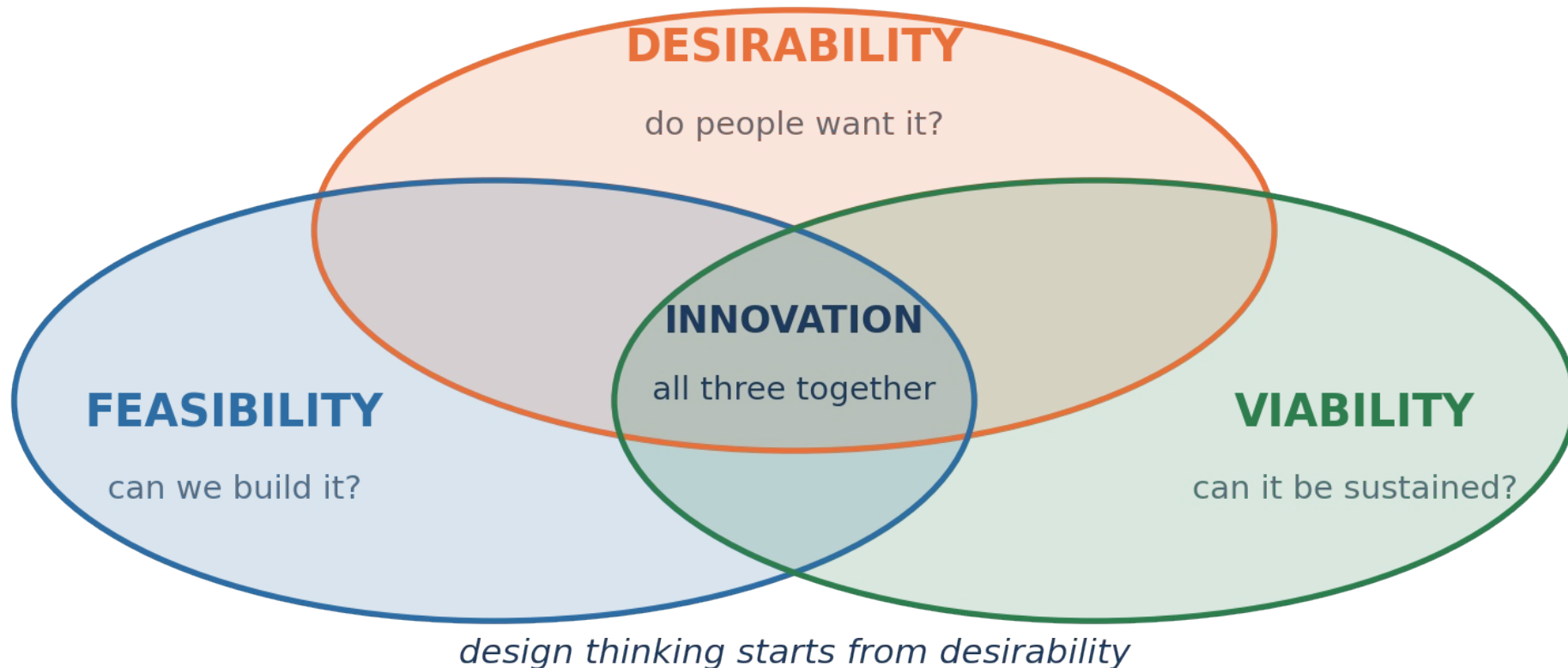
Starting from the people who will live with what you build

What “human-centred” means

- The people affected are the starting point of the design, not a test at the end
- Their needs are found by observing and listening, not assumed from your own experience
- They take part throughout – in framing, in choosing, and in testing
- Success is judged by what changes for them, not by what you were able to build

KEY POINT You are almost never the user. Designing for yourself is the commonest and costliest mistake.

The three lenses of innovation



An idea must survive all three – design thinking starts from desirability.

When one lens is missing

Desirable and feasible, but not viable

People love it and you can build it, but no one can afford to keep making it.

Desirable and viable, but not feasible

A promise that cannot be built with the technology and skills available.

Feasible and viable, but not desirable

A well-engineered, profitable product that nobody actually wants.

KEY POINT Feasibility and viability can be engineered later. Desirability cannot be added at the end.

Users, customers and stakeholders

The user

Uses the thing. Lives with every decision you make, including the ones you did not notice making.

The customer

Pays for it. Frequently not the same person as the user – a parent, a hospital, a department.

The stakeholder

Affected by it, or able to block it: a regulator, an installer, a family, a community.

KEY POINT Ask early who uses it, who pays for it, and who can stop it. They are rarely the same person.

Inclusive design and extreme users

- Design for the edges and the middle follows; design for the middle and the edges are excluded
- **Extreme users** – the expert and the complete novice – make hidden problems visible
- Age, literacy, language, disability, cost and climate are all design constraints
- A ramp built for a wheelchair is used by every trolley, pram and delivery cart

KEY POINT Accessibility is not a special case bolted on afterwards. It is ordinary good design.

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The engineer's role in society

Why this is part of an engineering course and not an optional extra

Engineering is a social activity

- Every artefact you design changes how somebody lives, works or travels
- Technical choices carry social consequences that no equation reports
- Who is included, who is excluded and who pays are design decisions, not afterthoughts
- India's largest problems – water, health, energy, mobility – are engineering problems with people at their centre

KEY POINT A design serves somebody. If you have not decided whom, the decision has still been made.

Where engineers meet society

Water and sanitation

Safe drinking water, treatment and reuse for households that have no reliable supply and no electricity.

Health and care

Affordable devices and clinics that work with the staff, power and transport that are actually available.

Energy and climate

Efficient machines, renewable generation and cooling that ordinary households can afford to run.

Mobility and safety

Transport and public space that are usable by women, children, older people and the disabled.

Professional and ethical responsibility

- You are accountable for foreseeable harm, including harm you did not intend
- Consent and privacy matter from the first user interview onwards
- Do not design to exploit a weakness – addictive loops and dark patterns are design failures, not features
- Say plainly what your solution cannot do; overclaiming is an engineering fault

KEY POINT “It was technically correct” has never been accepted as a defence.

Designing FOR and designing WITH

Designing FOR people

- Users are studied, then left out
- Experts decide what is needed
- Feedback comes at the end
- Solutions arrive from outside

Designing WITH people

- Users take part throughout
- They help frame the problem
- They test every prototype
- Solutions are built together

KEY POINT Communities that helped build a solution maintain it. Communities handed one often do not.

THINK ABOUT IT

Whose problem does your team's project actually solve – and how would you know if you were wrong?

- Who is the user, and have you met one?
- Who pays, and who can block it?
- Who is excluded by the way you have framed it?
- What would change for them if you succeeded?

Summary

- Human-centred means people are the starting point, not the final test
- Every idea must clear desirability, feasibility and viability
- Design thinking starts from desirability because it cannot be added later
- User, customer and stakeholder are three different people
- Designing for the edges produces better design for everyone
- Engineering decisions are social decisions, and carry ethical weight

Next: Case studies of successful innovations

- Embrace Infant Warmer – the reframe that changed the brief
- Aravind Eye Care and the Jaipur Foot
- MittiCool and Tata Swach
- What all five cases have in common